Clinical Decision Support Using NLP and Regular Expressions

DSA031 – Natural Language Processing

Guided by,
Dr. Gnanavel V K

Presented By,
G Uday Siva Sai(192425259)
T Bhanu Teja(192472259)

Introduction

- **Clinical Decision Support (CDS)** is an intelligent healthcare system that helps doctors make better clinical decisions.
- It uses **Natural Language Processing (NLP)** to understand and analyze medical text such as patient records, prescriptions, and doctor's notes.
- **Regular Expressions (Regex)** are used to identify and extract important information like patient IDs, dates, medicines, and laboratory values.
- This combination improves diagnosis accuracy, reduces medical errors, and enhances the quality of patient care.

Why NLP is Needed in Healthcare

Time Savings

Automate chart review tasks and surface critical findings to clinicians in seconds rather than minutes.

Patient Safety

Detect drug interactions, allergies, and critical lab trends to reduce preventable harm.

Analytics & Research

Scale cohort discovery, outcome measurement, and quality reporting across millions of notes.

Role of Regular Expressions (Regex)

- **Extracts important information** such as patient IDs, phone numbers, dates, and lab values from medical records.
- **Identifies medication names and dosages** quickly using predefined patterns.
- **Converts unstructured medical text** into organized and structured data.
- **Supports NLP** by providing fast and accurate pattern matching, helping doctors make better clinical decisions.

Engineer to Excel

NLP + Regex: Practical Workflow for Clinical Decision Support

- Regex extracts important details like patient IDs, dates, and medicines.
- NLP analyzes the extracted information to understand its meaning.
- The system organizes the data and provides alerts to help doctors make better decisions.

Data Collection

- Patient Records
- Doctor’s Notes
- Lab reports

Data preprocessing

- Remove Errors
- Standardize Dates
- Normalize Texts

Entity Extraction

- Diseases
- Medicines
- Allergies
- Symptoms

Context Analysis

- Identify Negations
- Check Medical History
- Understand Relationships

CDS

- Disease Alerts
- Treatment Suggestions
- Electronic Health Record(EHR)

Engineer to Excel

Regex + NLP Approach

- Combines **Regular Expressions (Regex)** and **NLP** for efficient medical text analysis.
- Extracts important information quickly and accurately from patient records.
- Reduces medical errors and supports better clinical decisions.
- Improves patient care by helping doctors make faster and informed decisions.

Scalable Automation

Process thousands of notes per hour while maintaining high precision for key fields.

Explainability

Rules provide traceable origins of extracted facts for clinical review and QA.

Faster Validation

Regex-driven checks accelerate creation of gold-standard labeled datasets for model training.

Interoperability

Outputs mapped to vocabularies (SNOMED, LOINC, RxNorm) enable downstream CDS and reporting.

Challenges

- **Data Heterogeneity**

  Different EHRs, templates, and clinician shorthand require adaptable rule-sets and continual model tuning.

- **Negation & Temporality**

  Identifying whether a condition is present, historical, or hypothetical remains challenging.

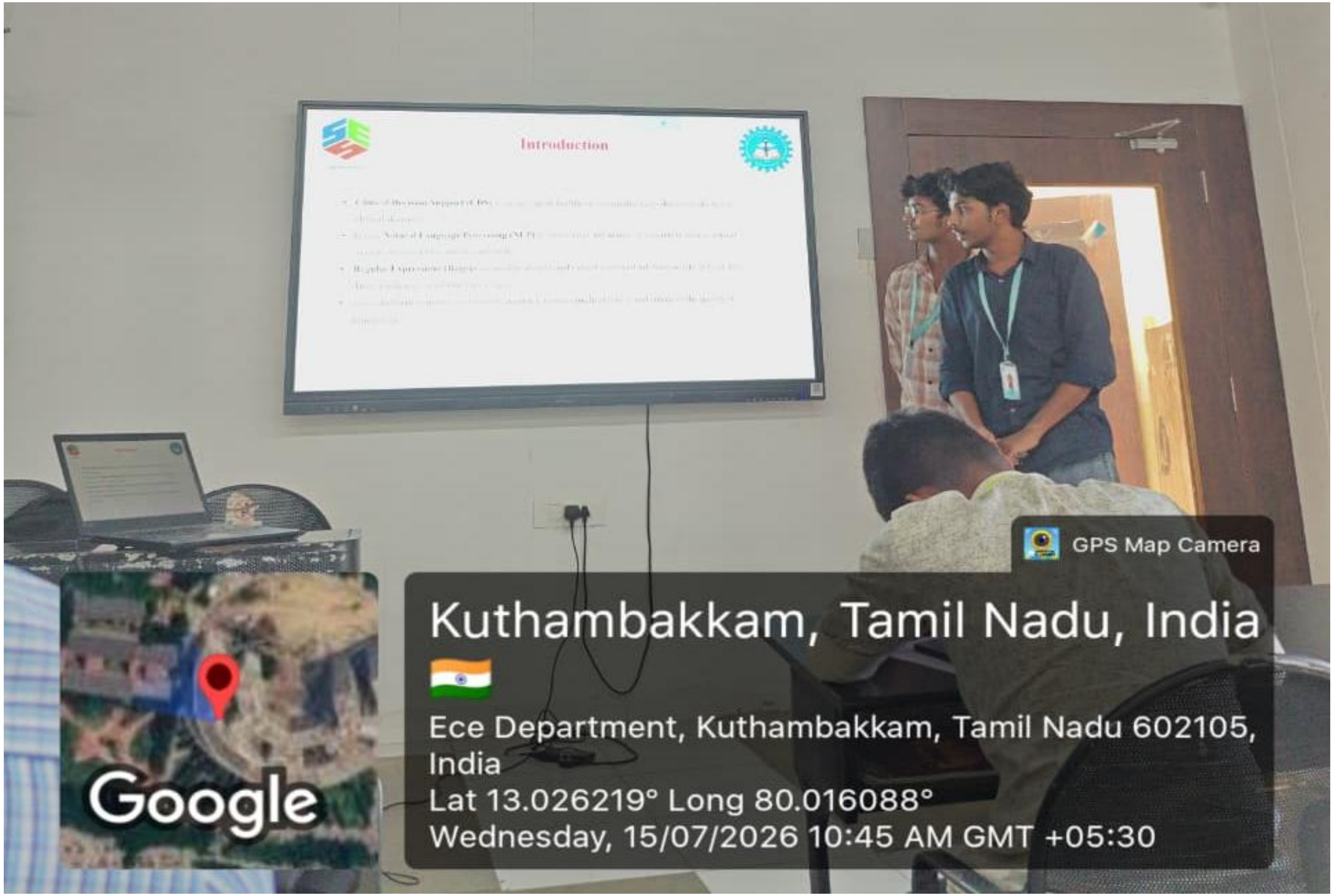
- **Clinical Validation**

  Prospective trials and clinician-in-the-loop testing are required to measure impact on outcomes and workflow.

- **Privacy & Deployment**

  On-premise or federated deployments, de-identification, and secure model governance must be prioritized.

Thank You


Introduction
GPS Map Camera
Kuthambakkam, Tamil Nadu, India
Ece Department, Kuthambakkam, Tamil Nadu 602105, India
Lat 13.026219° Long 80.016088°
Wednesday, 15/07/2026 10:45 AM GMT +05:30
Google